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Robotic Process Automation (RPA) and Virtual Reality Implementation in Engineering Education

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Abstract

Innovative technologies have the potential to revolutionize traditional learning methods, and engineering education is no exception. The chapter explores how integrating robotic process automation (RPA) and virtual reality (VR) enhances engineering education's experiential aspects by

creating immersive and interactive virtual laboratories. It aims to bridge the gap between theoretical knowledge and hands-on learning to understand complex engineering concepts better. Also, it presents a case study that demonstrates the impact of VR technology in engineering education, focusing on the components of a car engine. It aims to develop a virtual reality framework to train students in the engineering sectors, which will include learning assembly and maintenance of machine parts in a safe and interactive environment. It critically analyzes some of the previous studies about AR and VR systems in the training sector and education, which will help identify the gaps and problems in implementing actual training sessions in the engineering field. The questionnaire results show that 180 participants participated, 54% males and 46% females. 71% of the participants were aware of virtual reality technologies. The results also show that 64% agreed that VR training sessions could reduce the mechanical risks anticipated during practical learning, and only 16% disagreed.

***Keywords*:** Robotic process automation, virtual reality, modern education, virtual laboratories, student learning

5.1 Introduction

Robotic process automation (RPA) is a technology that uses software robots to automate repetitive and rule-based tasks. It can help businesses improve efficiency, reduce errors, and save time and money. RPA bots are widely adopted across industries and academic domains. It facilitates efficient collaboration analysis and deployment of RPA associations [1]. Robotic process automation (RPA) has become increasingly popular due to its ability to collect and integrate data with AI-based systems for analysis. This has made it an attractive solution for solving real-world problems across various industries and application domains [2]. Vulpe *et al.* (2023) examined the potential of introducing robotic process automation (RPA) in the education system [3]. They explored the various possibilities of implementing RPA and the sources of inspiration in line with the current fields where these software robots are already being used and are in high demand. Digital transformation at universities involves digitizing both administrative and academic processes. Robotic process automation

can help by automating processes quickly, accurately, and reliably through digital robots. At Izmir Bakircay University, a digital robot called ONDOscop was designed to automate scholarship payments, saving 95% time and cost per process execution with 100% accuracy [4]. Researchers and educators have been investigating strategies for providing creative educational experiences inside and outside the classroom [5]. The increased use of emerging technologies will have future value in supporting the creativity of modern teaching materials [6]. Emerging VR technology has proven great potential as a training platform for building education and learning [7]. Virtual reality technology is a valuable tool in training individuals in the construction industry and teaching life skills to children with autism. However, despite being widely validated, VR technology still has a low rate of acceptance and usage. This is partly due to the need to better understand users' attitudes toward its use [8]. By enabling dynamic training in a virtual environment, virtual reality technology can enhance the learner's interaction and collect training feedback [9]. Students can explore complex environments and situations through virtual immersion, manipulating environment elements to comprehend information. Many studies analyzed the effectiveness of virtual reality in higher education through a systematic literature review [10]. Virtual reality (VR) means "using computer technology to build a simulation environment" that could be artificial, for example, an animated scene or an actual place that was included in a virtual reality app. The implementation of AR/VR makes systems more accessible and minimizes manufacturing costs. They are also being utilized in training students, which is a task that comes with several issues and challenges, such as the costs of hardware and tools [11]. Today, augmented reality (AR) is widely used in training, production, maintenance, and manufacturing [12]. The virtual lab session is designed to provide students with an immersive learning experience beyond traditional teaching methods. With the VR environment, students can closely inspect each part of the engine to understand how it functions [13]. The virtual environment with interactive features enables them to disassemble and reassemble the engine virtually, while multimedia elements such as high-resolution pictures and instructional videos increase the visual experience. The VR lab goes beyond static images and videos by incorporating realistic simulations, allowing students to start the engine virtually, observe the movement of pistons, and witness the in-

teraction of various components. The proposed system helps evaluate the learning outcomes of students who participated in the VR lab session compared to traditional instructional methods. The results show that students who experienced the VR lab had increased engagement, improved understanding of engine components, and enhanced knowledge retention [14].

Previous studies identified several research gaps; addressing these gaps can provide valuable insights into the effective use of VR and RPA technologies to transform educational experiences and enhance student learning outcomes in learning and laboratory sessions. These gaps include the following:

- Integrate how VR and RPA are collaborative in learning environments. Investigate how VR and RPA technologies can seamlessly integrate into collaborative learning settings, such as group projects or team-based activities, to enhance student engagement and foster a strong sense of teamwork. Also, explore the effectiveness of incorporating VR simulations and RPA-driven automation tasks into interdisciplinary projects, where students from diverse fields can leverage their respective expertise to solve complex problems, encouraging a collaborative and motivated learning environment.
- Examine how the assessment methods of learning outcomes in VR-RPA. Develop robust assessment methods to measure the impact of VR and RPA technologies on student learning outcomes in laboratory sessions. Explore innovative assessment approaches, including cognitive skills, critical thinking abilities, problem-solving proficiency, and domain-specific knowledge acquisition.
- Explore innovative strategies for effective implementation.
- Identify best practices and curriculum strategies for effectively integrating VR and RPA into educational settings, including instructor training, curriculum design, and instructional delivery methods. Educators can investigate how to construct learning experiences and optimize student learning and engagement while utilizing VR simulations and RPA automation tools.
- Investigate long-term impact and sustainability.

Investigate the long-term impact of integrating VR and RPA technologies into educational curricula and lab sessions on student learning outcomes, career readiness, and retention in STEM fields. Assess the feasibility and sustainability of VR-RPA implementations in educational institutions, considering factors such as cost-effectiveness, infrastructure requirements, and faculty support.

5.2 Ethical Factors

Implementing virtual reality and robotic process automation in educational institutions presents several ethical considerations that need discussion and attention. An experimental implementation of a VR-RPA system using Unity software will be undertaken to create an immersive virtual environment where users can interact with RPA-driven automation tasks [15]. We will ensure informed consent from participants, protect their privacy and confidentiality, and adhere to ethical guidelines and regulations. The following are some key ethics issues related to their implementation:

- Security and safety standards:
  The VR-RPA project should follow safety and security standards and ensure that it does not cause any unacceptable risk of injury or physical harm. One of the project's main objectives is to develop a safe environment that enables adequate security and safety [7]. Therefore, it has no negative impact on the user because the project aims to design and develop a virtual environment that is safe and secure for everyone.
- Data privacy:
  VR-RPA systems may collect sensitive data about users' behaviors, interactions, and physiological responses. It is crucial to protect student privacy, and data collection is conducted transparently and with informed consent. In this project, we developed a questionnaire to collect some data required for this project, which required maintaining privacy and the responsibilities associated with collecting and using personal information. Also, it allows people to understand what data is being collected and how it is used.
- Intellectual property:

The VR-RPA project should not apply any experiments that are incompatible with the ethics concepts. It must preserve intellectual property and mention all the sources used in this project. So, it should use original crated activities, such as designs, symbols, names, and images. The project should protect the patents, copyrights, and trademarks of others and make sure that it applies the ethics rules used in every practical application.

- Public interest:
  Implementing a VR-RPA project should consider the public interest (user, Clint, etc.) as one of our objectives to serve and maintain public benefit and social care. The project leader must accept full responsibility for the project and approve a high-quality design that is safe (life, environment, and users) with no associated harm. They must also cooperate to address any concerns the project, support, or documentation will cause.
- Bias and fairness:
  Institutions must consider the ethical implications of deploying VR-RPA systems in educational settings, including the potential impact on employment opportunities, professional development, and the human-centered aspects of teaching and learning [16].

5.3 Research Design

We aim to determine the appropriate research design and methodology based on the research objectives and questions. This research performs both quantitative and qualitative approaches [17]. The following steps are implemented to design the proposed VR-RPA system. Moving forward, we aim to develop and design a comprehensive virtual reality framework tailored to engineering sectors. This framework will provide students with interactive training modules for learning the assembly and maintenance of machine parts in a safe and engaging environment. Additionally, we will conduct critical analyses of previous studies on AR and VR systems in training and education sectors to identify gaps and challenges in implementing practical training sessions in engineering fields.

1. The project starts with a literature survey of previous studies to investigate the impact of VR-RPA integration on educational outcomes and

explore user experiences, perceptions, and attitudes toward these technologies and research gaps.

2. The second step is to choose appropriate data collection methods to gather relevant information and insights related to VR-RPA implementation in education.
3. Analyze the collected data using appropriate analytical techniques and methods in SPSS.
4. Ensure ethical conduct throughout the research process by obtaining informed consent from participants, protecting their privacy and confidentiality, and adhering to ethical guidelines and regulations.
5. An experimental implementation of a virtual reality (VR) and robotic process automation (RPA) system using Unity software involves creating a virtual environment where users can interact with RPA-driven automation tasks in an immersive VR setting.

Figure 5.1 shows the schematic diagram of the System design, which provides a visual representation of how the various components of the system interact and function together. The virtual environment encloses the 3D design, where users interact and navigate the proposed system while wearing the VR headset. This environment may include virtual objects, structures, and user interfaces rendered to create an immersive experience and interact with the virtual environment using VR headsets and controllers, engaging with educational content, simulations, or RPA-driven tasks.

5.4 Research Questions

This section presents research questions to examine the impacts of virtual reality (VR) technology on thinking skills, mental skills improvement, social environments, and teacher-learner interactions. The chapter aims to understand better how VR technology impacts cognitive skills, such as attention, memory, and problem-solving abilities, as well as mental health factors, including stress, anxiety, and depression. Additionally, it explores how VR technology can influence social environments, such as communication, relationships, and social skills, and how it can enhance teacher-learner interactions, including engagement, collaboration, and feedback. By answering these research questions, we hope to gain valuable insights

into the potential of VR technology to impact various aspects of human life positively.

- Virtual Reality and Thinking Skills:
  How does VR technology affect students' cognitive thinking skills?
- Mental Skills Improvement through VR:
  How does VR enhance cognitive skills like focus, attention, and memory?
- Social Learning Environment with VR:
  Can VR technology enhance social and communication skills?
- Teachers and Learners Interaction Using VR:
  How can VR technology enhance interaction between teachers and learners?

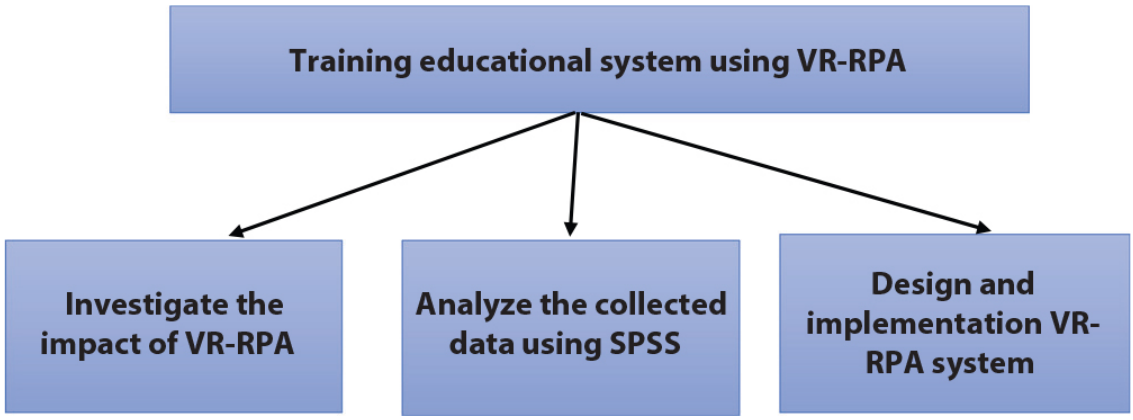


Figure 5.1 Schematic diagram of the proposed system design.

This chapter aims to explore these classifications in detail and their benefits while reviewing their main advantages.

5.5 Experimental Design

The Unity Hub, a versatile centralized management tool, is a hub for developers, VR enthusiasts, and educators alike. It allows for the organization and creation of a wide range of 2D and 3D project capabilities, including games, simulations, and VR-RPA applications. The Unity Hub is not just a tool but a partner in the development workflow for VR projects. It provides a streamlined interface for accessing VR development tools, plugins, and resources, catering to the diverse needs of its users [18]. A set of productive virtual reality tools enables users to create VR and AR

applications quickly and naturally through a simple interface, making it easy to interact with the virtual environment using a VR-easy plugin.

In addition, it supports major head-mounted display (HMD) devices, ensuring compatibility and versatility. We also use the JavaScript programming language, which enables us to execute complex features on web pages. With JavaScript, we can display videos, pictures, content updates on the webpage, interactive maps, and 2D/3D animation. Figure 5.2 showcases the design of the proposed virtual lab using Unity, a testament to the versatility and power of the Unity Hub.

Multiple educational rooms are depicted within the virtual environment, each designed to facilitate specific learning activities related to the required learning outcomes [19]. These rooms are accessible to VR headset users and can be navigated through interactive interfaces or teleportation mechanisms. Each educational room is designed to provide immersive and interactive learning experiences that complement traditional classroom instruction. Users can explore the virtual environment at their own pace, interact with educational content, and engage in experiential learning activities that reinforce key concepts and skills [20].



Figure 5.2 The proposed virtual lab using unity hub.

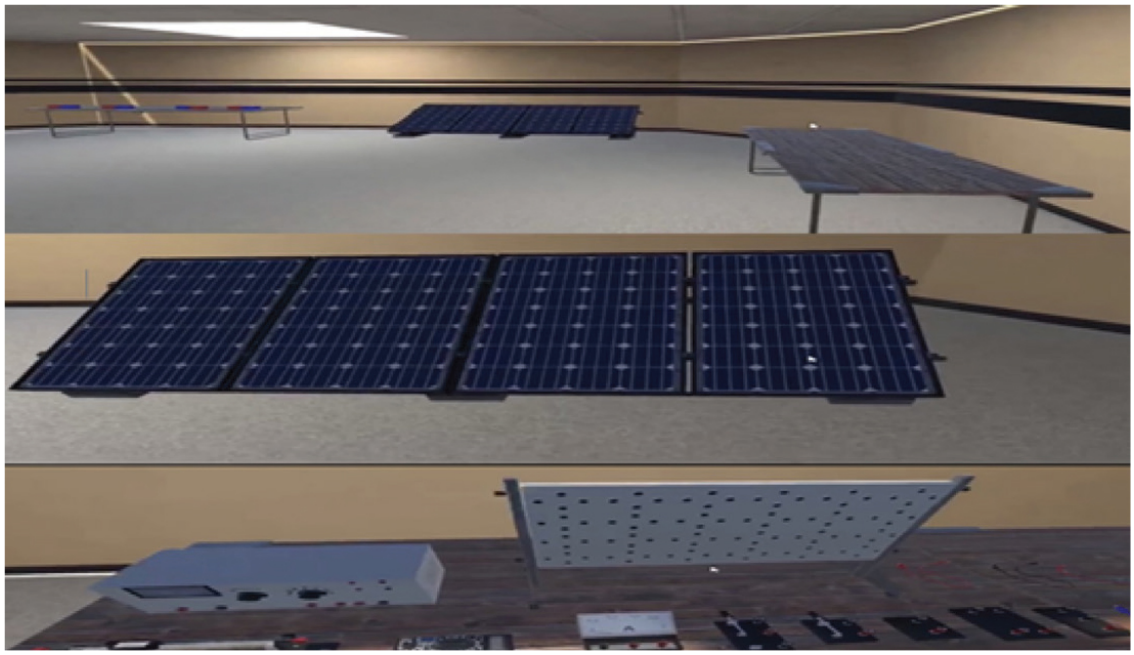


Figure 5.3 Solar panel systems and electrical lab sets.

The first educational room is dedicated to solar panel and electrical lab activities. It contains a realistic simulation of a solar panel setup along with an electrical lab set, allowing students to learn how to install, configure, and troubleshoot solar panel systems, as presented in Figure 5.3.

This activity aligns with learning objectives related to renewable energy, electrical engineering, and sustainability. The next room focuses on materials and activities related to car engines, as shown in Figure 5.4. It features interactive displays showcasing various car engine components, including pistons, valves, and cylinders. Additionally, educational videos

provide instruction on engine maintenance, repair, and troubleshooting techniques. Students can interact with activities using hands-on engine parts, such as dismantling and reinstalling parts within the virtual environment, providing practical experience and enhancing their understanding of automotive engineering concepts.



Figure 5.4 Car engine components.

5.6 Results and Discussion

This research performs both quantitative and qualitative approaches. A well-designed questionnaire is developed and distributed online to gather needed data for implementing VR-RPA in the teaching environment. The collected data will be used to examine the impact of VR-PRA on enhancing teaching performance. The questionnaire targets students in engineering institutions to discuss how they are familiar with virtual reality and its effects on improving education. SPSS software was used to analyze the questionnaire. Table 5.1 shows the descriptive statistics and demographic details of the participants. It is shown that a total of 180 participants participated in the questionnaire.

Figure 5.5(a) depicts the gender distribution of participants in a survey on the use of virtual reality technology in Oman. It reveals that 54% of respondents were male, while females accounted for the remaining 46%. Consequently, there was a higher representation of male participants compared to females across various specialties. Figure 5.5(b) illustrates that the highest percentage of participants in the study were between 22

and 24 years old, making up 44% of the total. Additionally, respondents aged 25 to 27 constituted the second-largest group, comprising 28% of the interviewees. Following this, individuals aged 19 to 21 accounted for 17% of the total participants, while those above 28 years formed the smallest segment in the survey. Figure 5.5(c) depicts the employment status of participants, which shows that 68% of interviewees were not employed, while the remaining 32% belonged to the employed category. Figure 5.5(d) indicates that most respondents had a specialization in mechanical engineering, comprising 68% of the total. Additionally, 12% specialized in chemical engineering. Following this, 11% of participants had specializations in electrical engineering, 7% in civil engineering, and 1% each in chemical & electrical engineering and English studies. Figure 5.6 presents the respondents' level of awareness regarding virtual reality (VR). The data reveals that a significant majority, comprising 71% of the participants, demonstrated awareness of VR technology. Conversely, 29% of respondents indicated they were unaware of VR. This suggests a noteworthy level of familiarity with VR among the surveyed population.

According to Figure 5.7, most respondents strongly agreed or agreed that virtual reality promotes better memory retention of lessons and makes it easier to engage students consistently. Expressly, 41% strongly agreed, while an additional 26% agreed with this statement. Table 5.2 shows the statistics results of the participants.

Table 5.1 Descriptive statistics and demographic details of the participants.

	N	Min	Max	Mean	Std. deviation	Variance
Gender (male, female)	180	1	2	1.54	0.500	0.250
Marital status (single, married)	180	1	2	1.27	0.443	0.197
Working status (yes, no)	180	1	2	1.32	0.469	0.220
Major/specialization	180	1	7	1.69	1.196	1.431
Level (1, 2, 3, 4)	180	1	4	3.07	0.919	0.845
Residence (names of governorates)	180	1	24	9.08	5.593	31.284
Do you know virtual reality? (yes, no)	180	1	2	1.29	0.457	0.209

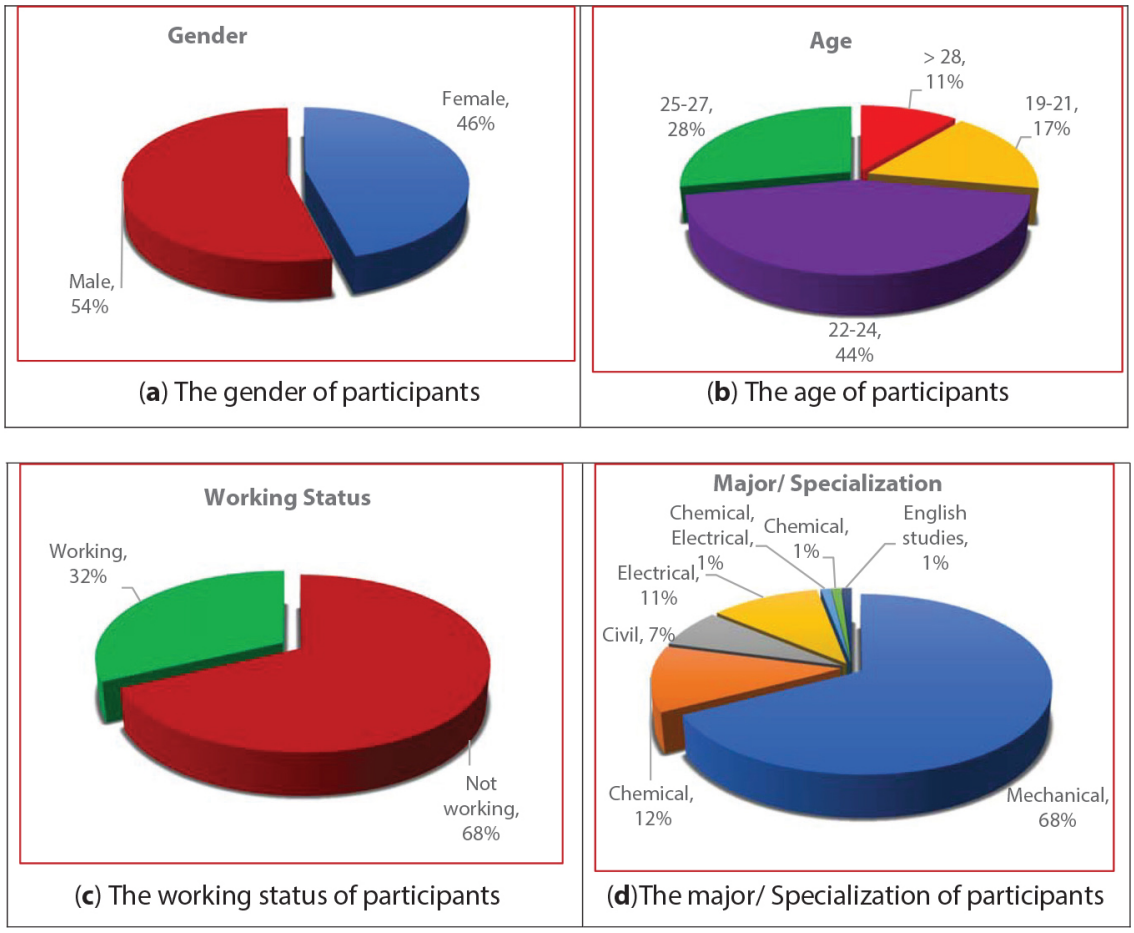


Figure 5.5 Descriptive statistics demographic details of the participants (a, b, c, and d).



Figure 5.6 Answer of "How the participants are familiar with virtual reality."

Figure 5.8 presents the participant's views regarding the impact of virtual reality technology on the development of thinking skills. According to the chart, most participants agreed with this notion, with 33% strongly agreeing and 32% agreeing. However, 23% of interviewees remained neutral, while 10% disagreed and 2% strongly disagreed that virtual reality technology contributes to developing thinking skills. Additionally, some re-

spondents provided qualitative feedback, stating that the virtual reality experience provides them with a general idea of practical sessions. When students repeat lab exercises and review different solutions, they improve their thinking skills and problem-solving abilities. This iterative process enhances their abilities and boosts their confidence.

Table 5.2 Statistics results of the participants answers.

Question	N	Min.	Max.	Mean	Std. deviation	Variance
Have you ever used virtual reality before	180	1	2	1.99	0.075	0.006
Virtual reality techniques make easy to understand subjects	180	1	5	3.77	1.201	1.442
Easy to remember lessons when Virtual reality is used	180	1	5	3.86	1.223	1.495
Study time becomes less using Virtual reality	180	1	5	3.82	1.154	1.331
Virtual reality technology develops thinking skills	180	1	5	3.86	1.047	1.096
Engineering designing process and architectural	180	1	5	3.98	1.046	1.094

Question	N	Min.	Max.	Mean	Std. deviation	Variance
lessons can be made simpler						
Mental skills can be improved	180	1	5	3.92	1.080	1.167
Using virtual reality students may lose their originality of thinking	180	1	5	3.78	1.234	1.523
Virtual reality education gives the opportunity to make learning environment social by allowing students to communicate with each other	180	1	5	3.84	1.098	1.205
Virtual reality allows bridging the gap between teachers and	180	1	5	3.98	1.073	1.150

Question	N	Min.	Max.	Mean	Std. deviation	Variance
learners making the student more confident						
Virtual reality technologies allows users to see and live with virtual environments and objects	180	1	5	3.94	0.998	0.996
Virtual reality will change the role of a teacher from content delivery to content facilitation	180	1	5	3.95	1.058	1.120
Virtual reality in education makes it easy to engage students all the time, making experiences memorable	180	1	5	3.73	1.096	1.202

Question	N	Min.	Max.	Mean	Std. deviation	Variance
Valid N (listwise)	180					

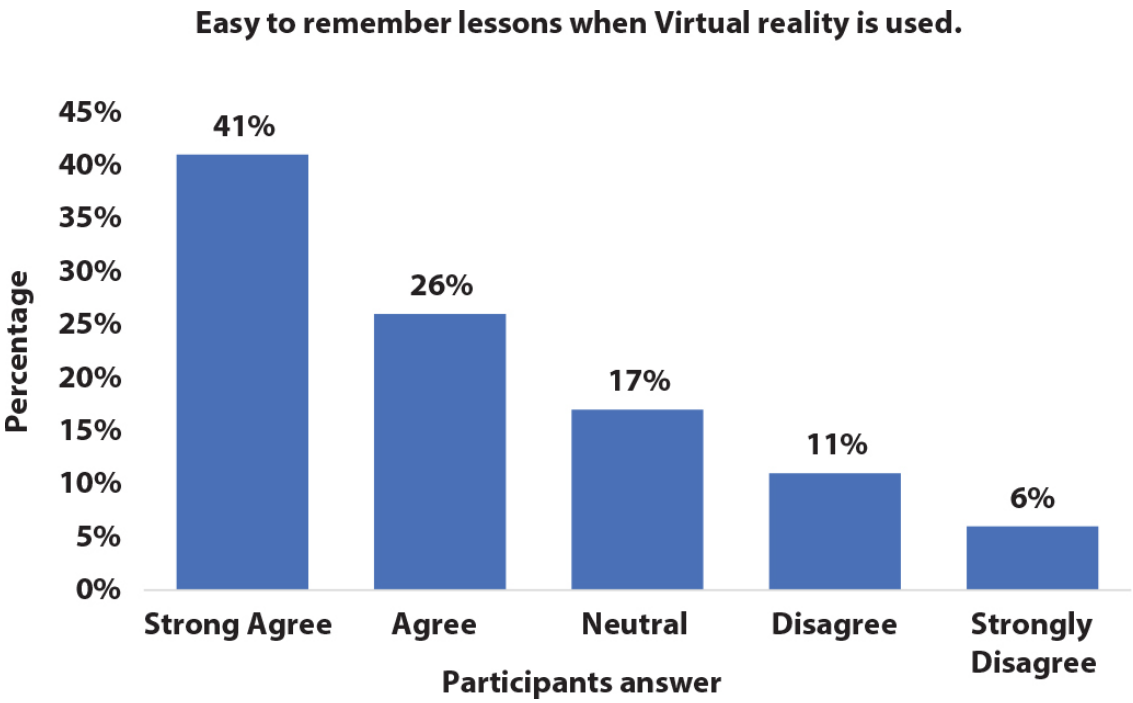


Figure 5.7 Answer of "Easy to remember lessons when VR is used (N 180)."

Figure 5.9 illustrates that 34% of the respondents strongly agreed that virtual reality education provides an opportunity to foster social learning environments, enabling students to communicate with each other. While 31% agreed with this statement, the data reveals that 22% of respondents maintained a neutral stance. Moreover, the qualitative feedback from respondents sheds further light on their perspectives. Many respondents highlighted the immersive nature of virtual reality experiences, emphasizing how they enable collaborative learning activities and peer interactions. They noted that VR environments facilitate realtime communication and collaboration among students, allowing them to collaborate on projects, discuss concepts, and share ideas irrespective of physical distance.

Figure 5.10 presents the responses of participants regarding the assertion that virtual reality enhances mental skills. Approximately 38% of participants strongly agreed that virtual reality education enhances mental

skills, while 34% agreed that mental skills are improved through VR experiences. However, a significant portion, constituting 16% of the interviewees, remained neutral about the claim. Furthermore, qualitative feedback provided more profound insights into participants' perspectives. Many participants described how virtual reality immerses them in engaging environments that challenge their cognitive abilities, thereby refining their mental skills. Additionally, some respondents highlighted the potential therapeutic benefits of VR for addressing mental health conditions such as test anxiety and utilizing specialized stress reduction techniques. Conversely, approximately 12% of respondents disagreed or strongly disagreed that VR would enhance mental skills, expressing doubt about the effectiveness of VR-based interventions in improving cognitive functioning.

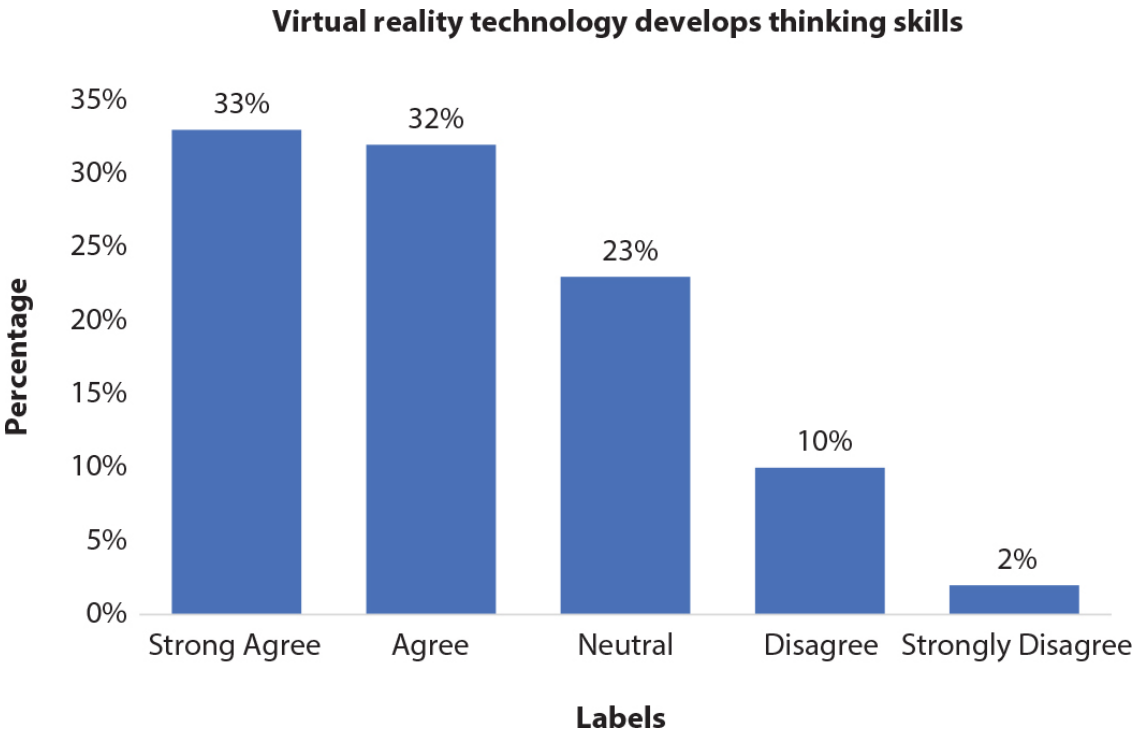


Figure 5.8 Answer of "virtual reality technology develops thinking skills."

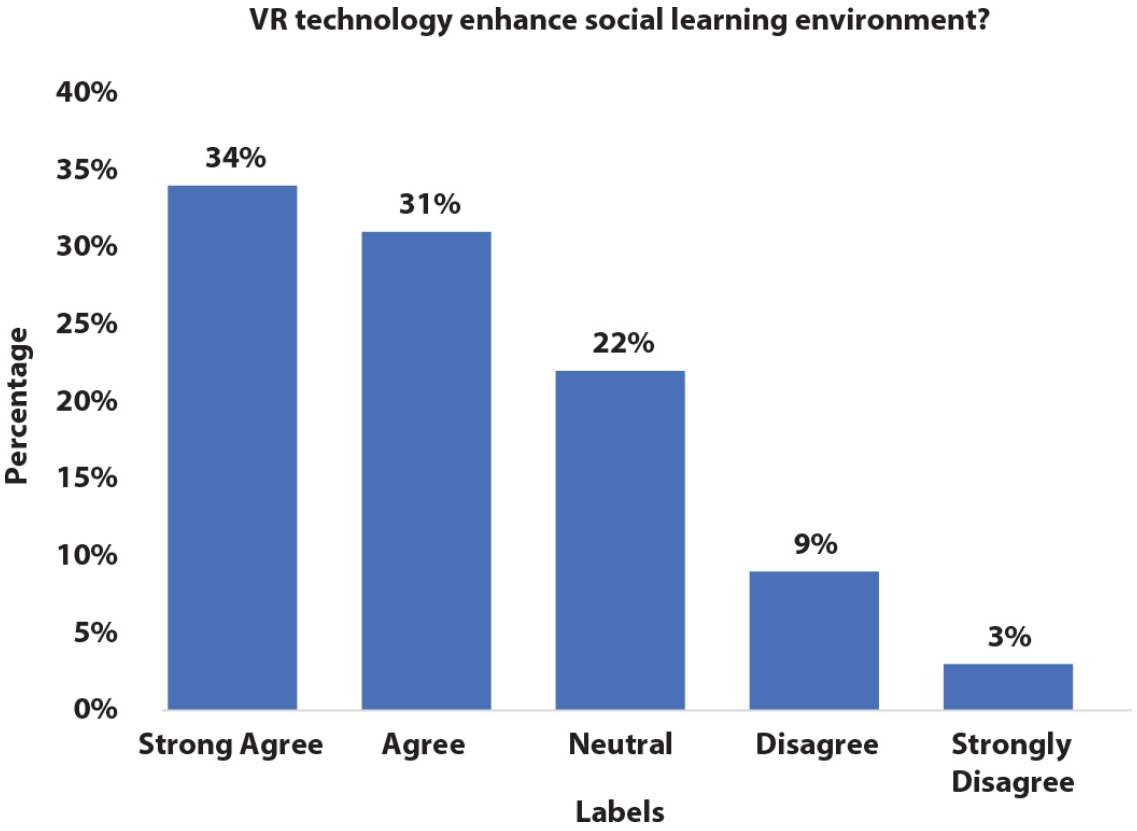


Figure 5.9 Answer of "VR technology enhance social learning environment."

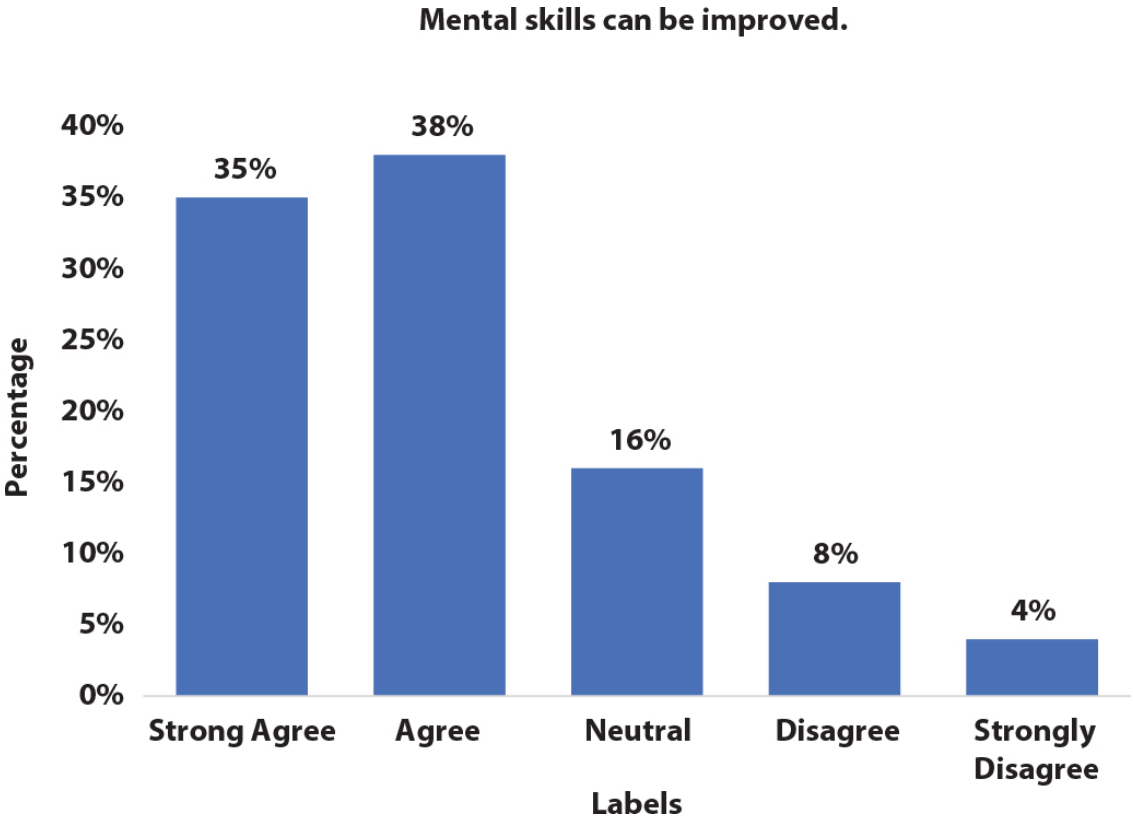


Figure 5.10 Answer of "mental skills can be improved."

In Figure 5.11, around 39% of the respondents strongly agreed that virtual reality facilitates bridging the gap between teachers and learners, thereby boosting student confidence. An additional 32% expressed their agreement, while 18% remained neutral. On the contrary, 7% disagreed, and 3% strongly disagreed with the notion that virtual reality aids in bridging this gap and fostering student confidence. Furthermore, qualitative feedback elaborated that virtual reality facilitates personalized and interactive learning experiences, allowing for more effective communication and engagement between teachers and students. They described how VR technology enables educators to create immersive and dynamic lessons that cater to individual learning styles and preferences, thereby building student confidence and motivation.

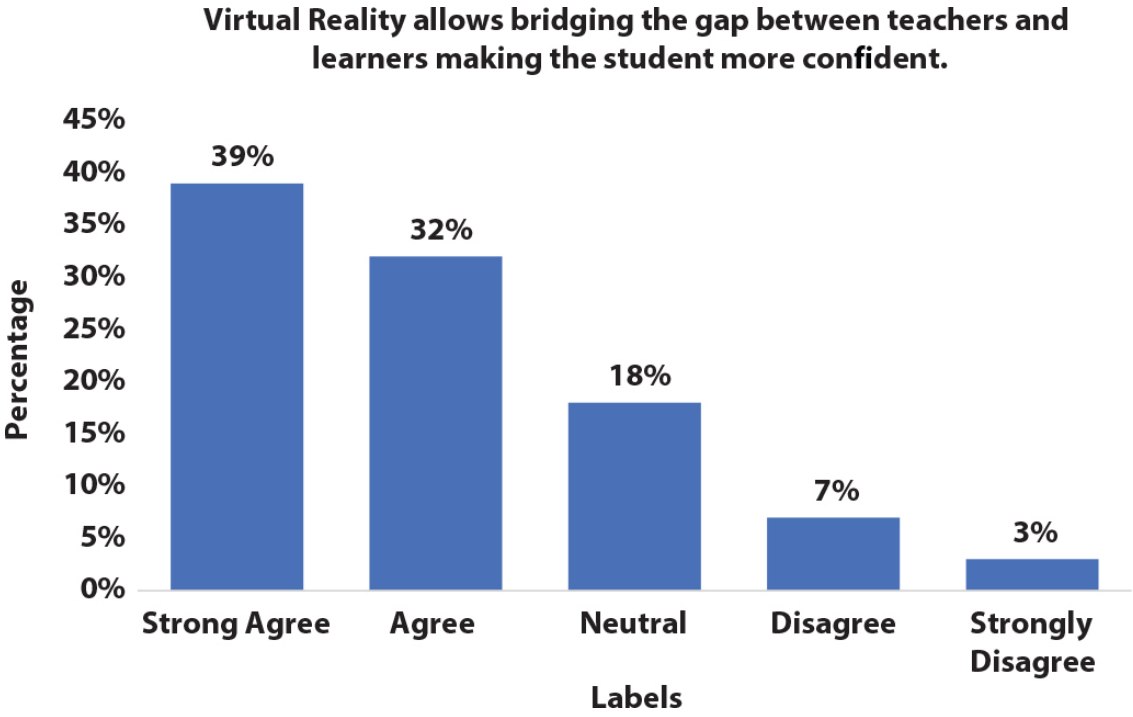


Figure 5.11 Answer of "VR allows bridging the gap between teachers and learners."

5.7 Conclusion

Integrating robotic process automation (RPA) and virtual reality (VR) holds enormous promise for improving engineering education. Creating immersive virtual laboratories will bridge the gap between theoretical and practical learning, offering students a deeper understanding of complex engineering concepts. Through a case study focusing on engineering concepts, such as car engine components, we have seen firsthand how VR technology can enhance experiential learning in engineering education. Multiple educational rooms are depicted within the virtual environment, each designed to facilitate specific learning activities related to the required learning outcomes. These rooms are accessible to VR headset users and can be navigated through interactive interfaces or teleportation mechanisms. Each educational room is designed to provide immersive and interactive learning experiences that complement traditional classroom instruction. Users can explore the virtual environment at their own pace, interact with educational content, and engage in experiential learning activities that reinforce key concepts and skills. The questionnaire results, with 180 participants, reveal promising insights. About 54% of participants were male, and 46% were female. Most participants were aware of VR technologies, and a significant percentage agreed that VR training sessions could mitigate mechanical risks during practical learn-

ing and positively impact various aspects of engineering education. The survey reveals that most respondents specialized in mechanical engineering, comprising 68%. Additionally, 12% specialized in chemical engineering, 11% in electrical engineering, 7% in civil engineering, and 1% each in chemical and electrical engineering. Most respondents strongly agreed or agreed that virtual reality promotes better memory retention of lessons and makes it easier to engage students consistently. 41% strongly agreed, while 26% agreed with this statement. Also, 65% strongly agreed or agreed that virtual reality technology contributes to developing thinking skills. Additionally, 65% of the respondents strongly agreed or agreed that virtual reality education provides an opportunity to foster social learning environments, enabling students to communicate with each other. Approximately 72% of participants strongly agreed or agreed that virtual reality education enhances mental skills. Lastly, around 71% of the respondents strongly agreed and agreed that virtual reality facilitates bridging the gap between teachers and learners, thereby boosting student confidence. confidence.

Integrating the RPA-VR system holds significant promise for enhancing engineering education, but there are several limitations and challenges to consider:

- Cost: Developing and implementing immersive virtual laboratories can be expensive, requiring investment in VR hardware, software development, and ongoing maintenance.
- Technical Complexity: Integrating RPA and VR technologies requires expertise in both fields, as well as collaboration between engineers, educators, and software developers. Managing the technical complexity of these systems and ensuring seamless integration can pose challenges for implementation.
- Social Interaction: VR may offer simulated environments for interaction but could lack the depth and richness of face-to-face communication and interpersonal connections.
- Teacher-Student Interactions: While VR technology has been lauded for its potential to enhance teacher-student interactions, some doubt was expressed about its ability to replace traditional teaching methods or establish genuine rapport between teachers and students.

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Note

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